



GOVT. COLLEGE OF EDUCATION

Block 15, F. B. Area, Karachi.

Internal Examinations 2023

General Mathematics 342

B. Ed. (Hon.)

Semester II

Time: 2 Hours

Total Marks: 60

Section A (10 Marks)

(MCQs)

Q1. Choose the correct answer:

i. $(-1, -2)$ lies in _____ quadrant

(a) 1st (b) 2nd (c) 3rd (d) 4th

ii. 20% of 10 is _____

(a) 2.3 (b) 2.5 (c) 3 (d) 2

iii. Mean proportion between 2 and 8 is _____

(a) 16 (b) 32 (c) 256 (d) 4

iv. _____ is greatly affected by extreme values in the data

(a) Mode (b) Arithmetic mean (c) median (d) Variance

v. $1+2x^2+3xy^2$ having _____ degree polynomial

(a) 1 (b) 2 (c) 3 (d) 0

vi. Area of circle having radius 7cm is

(a) 49 cm (b) 22 cm² (c) 154cm² (d) 154mm²

vii. The angles less than 90° are called _____ angle.

(a) obtuse (b) Acute (c) Complementary (d) none of these

viii. Sum of angles in quadrilateral is _____

(a) 540° (b) 180° (c) 720° (d) 360°

ix. A _____ is a straight path of points that has no beginning or end

(a) ray (b) Line segment (c) Line (d) point

x. A matrix having determinant zero is called _____ matrix

(a) Zero (b) null (c) both a and b (d) Singular



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Section B (20 Marks) (Short Answer's Questions)

Q-02 Attempt any four of the following.

- Find the area of the triangle whose sides are 10cm, 13 cm, 15 cm.
- Following are the marks obtained by 10 students. Find the median 35, 67, and 30, 52, 57, 49, 66, 76, 22, 36.
- Find the distance and slope using (6,11), (10,5)
- Find the value of $\frac{4x^2y(z-1)}{a+b-3c}$
When $x = 2$, $y = -1$, $z = 3$, $a = 0$, $b = 4$, $c = 1/3$
- Write the formulae of area of any six quadrilateral.

Section C (30 Marks) (Detailed Answer's Questions)

Q-03 attempt any two of the following.

- (a) Prove that the vertices (7, 10), (4, 5), (10, 15) are of isosceles triangle.
(b) What should be added to $4a^4 + 4a^3 + 5a^2 + 2a + 5$ to make it a perfect square?
- Factorize the following
 - $a^2b^2 - 6ab + 9$
 - $1 + 4b^4$
 - $6a^2 + 11a - 10$
- (a) Solve if possible by using Cramer's Rule.
 $4x + y = 2$
 $7x + 2y = 3$
(b) Solve by using quadratic formula
 $k^2 - 2k - 63 = 0$